Types of Operating Systems

Learning Outcomes

- To Classify different types of Operating Systems.
- Understanding the working of Batch, Multiprogramming, Multitasking, Multiprocessing, Time-Sharing, Distributed, Network, Embedded, and Mobile Operating Systems.
- Comparing different operating systems based on features and applications.
- Identifying real-life examples of each type.

WINDOWS



WINDOWS
(DESKTOP/LAPTOP)

LINUX



LINUX
(SERVER/DEV)

ANDROID



ANDROID
(MOBILE PHONE)

MACOS



MACOS
(APPLE DESKTOP/LAPTOP)

SERVER



SERVER OS
(DATACENTER)

ATM



ATM OS
(EMBEDDED SYSTEM)

SMARTWATCH



SMARTWATCH
(WEARATCH (WEARABLE OS)

Why Different Types of Operating Systems?

Different devices have different requirements.

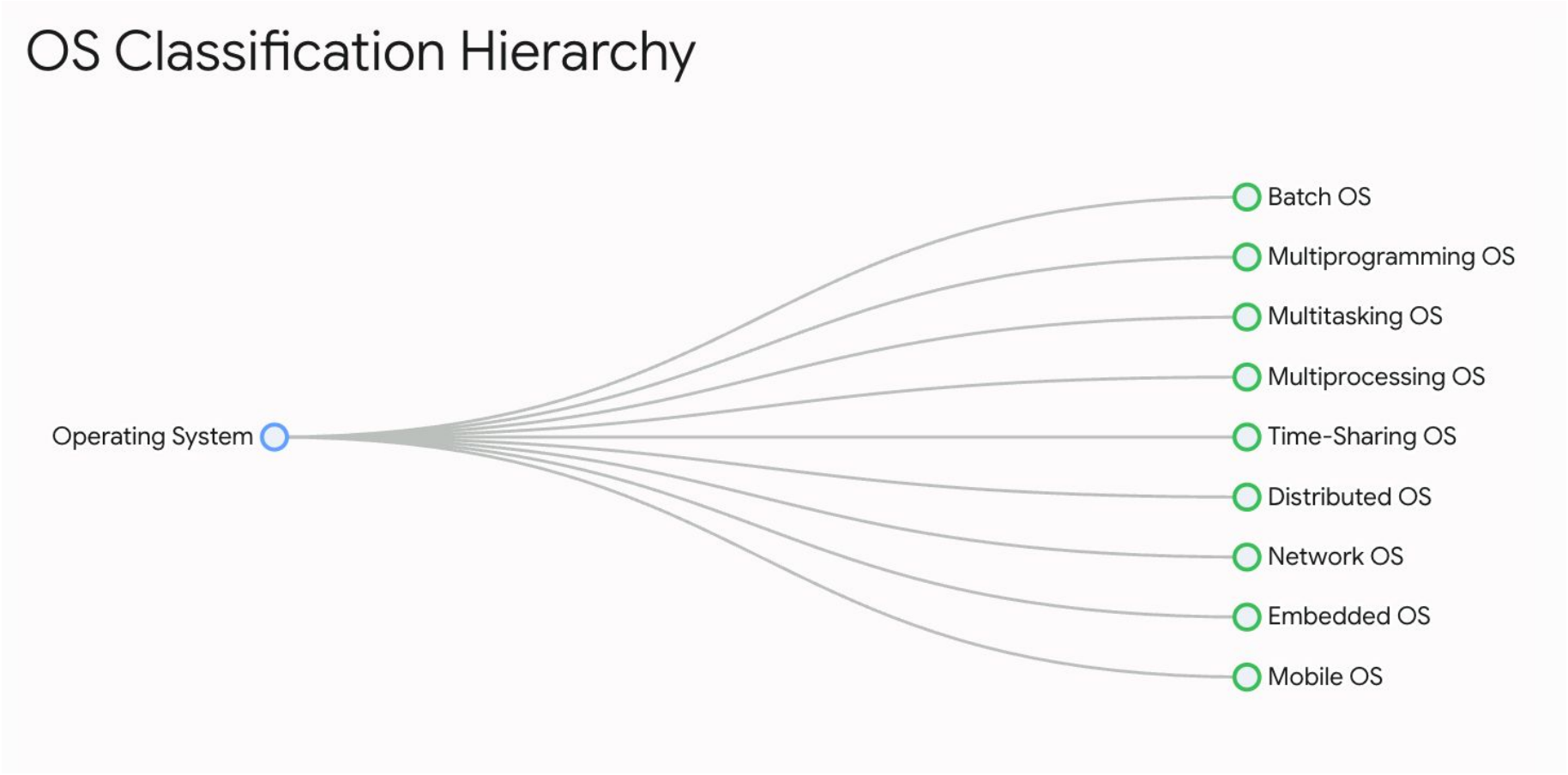
Examples:

- ATM Machine
- Laptop
- Mobile Phone
- Server
- Airplane
- Smart TV
- Industrial Robot

One Operating System cannot efficiently satisfy all these needs.

“Think about an ATM and your smartphone. Both contain processors and memory, but their tasks are very different. That’s why different operating systems are designed for different environments.”

Classification of Operating Systems



Core Classifications Summary

- **Batch OS:** Executes batches of similar jobs without direct user interaction during execution (e.g., payroll processing).
- **Multiprogramming OS:** Keeps the CPU continuously active by keeping multiple jobs in memory and switching to another when one waits for I/O.
- **Multitasking OS:** Uses rapid time-slicing to allow users to run multiple applications concurrently with real-time responsiveness.
- **Multiprocessing OS:** Leverages two or more physical CPUs or cores working concurrently to increase compute throughput.
- **Time-Sharing OS:** Allocates small slices of CPU time to multiple users logged into a single central system simultaneously.
- **Distributed OS:** Coordinates independent, networked physical computers so they appear to the user as a single unified system.
- **Network OS:** Runs on a server to manage network storage, security, files, and users across connected client machines.
- **Embedded OS:** Dedicated, lightweight system engineered to control specific embedded hardware with deterministic performance.
- **Mobile OS:** Optimized for handheld touchscreen devices, emphasizing power efficiency, wireless connectivity, and app sandboxing.

Batch Operating System

Definition

A Batch Operating System executes a group (batch) of jobs together without user interaction.

Characteristics

- No direct user interaction
- Jobs are grouped
- Automatic execution
- High throughput

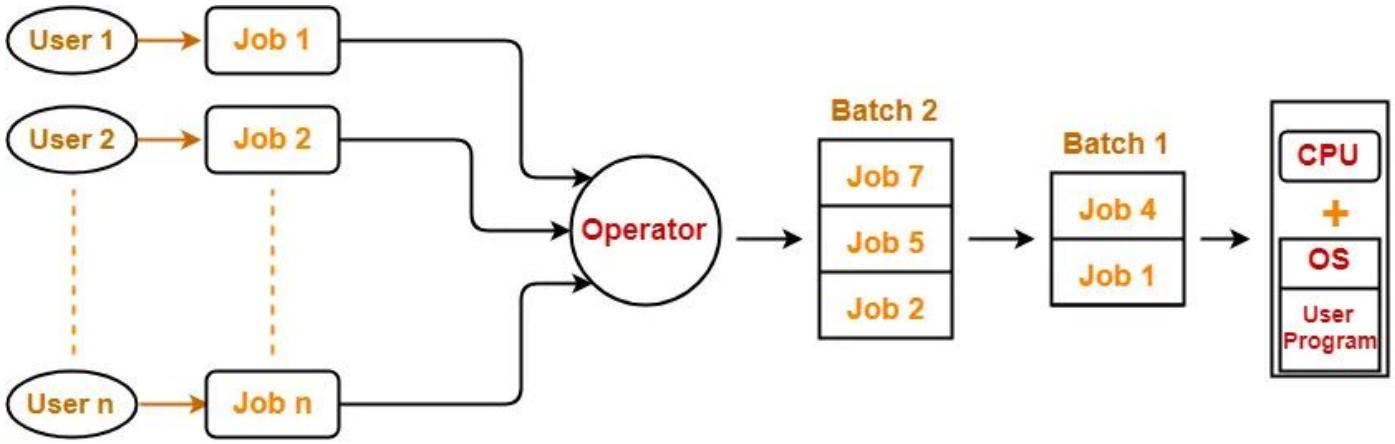
Examples:

- IBM Mainframe Systems

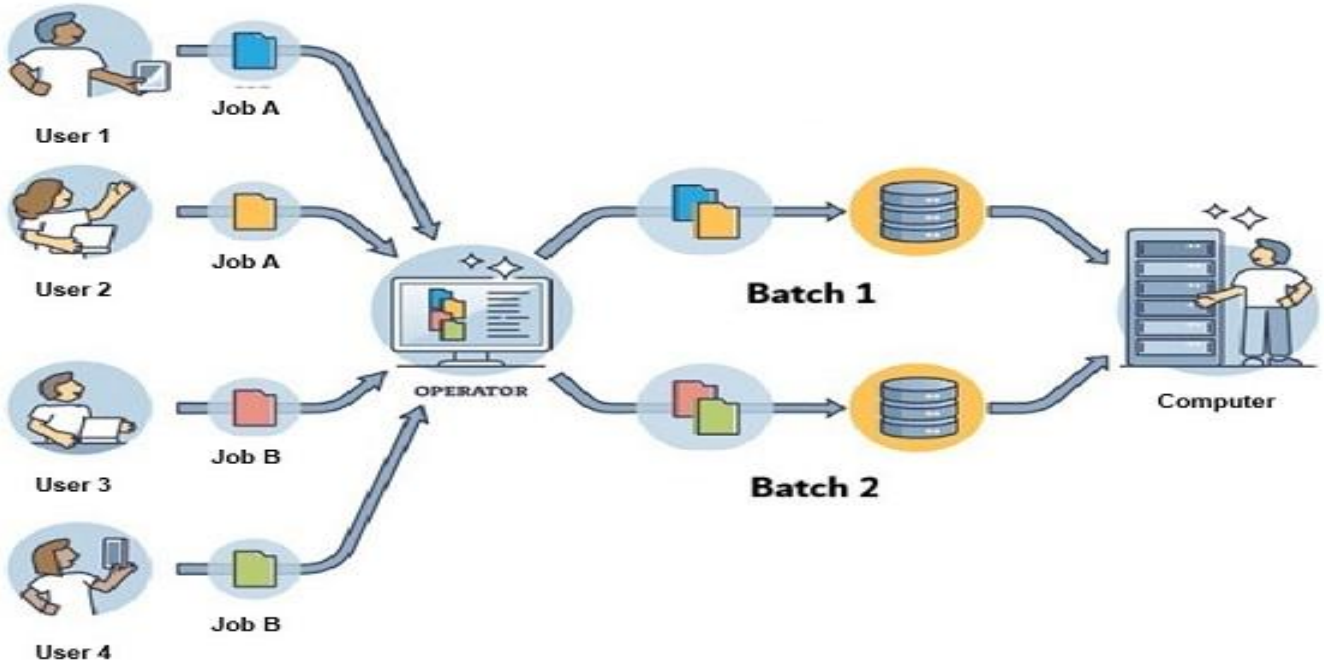
Applications:

- Payroll processing
- Bank statement generation
- Electricity bill generation

“Imagine a university processing 10,000 student result sheets overnight. Instead of processing one by one, all jobs are collected into a batch and executed automatically.”



Batch Operating System



Advantages and Disadvantages of Batch OS

Advantages

- High throughput
- Efficient for repetitive tasks
- Less idle CPU time

Disadvantages

- Long waiting time
- Difficult to debug
- No interaction with users

Multiprogramming Operating System

Definition

Multiple programs are kept in memory simultaneously, and the CPU switches to another program whenever one is waiting for I/O.

Features

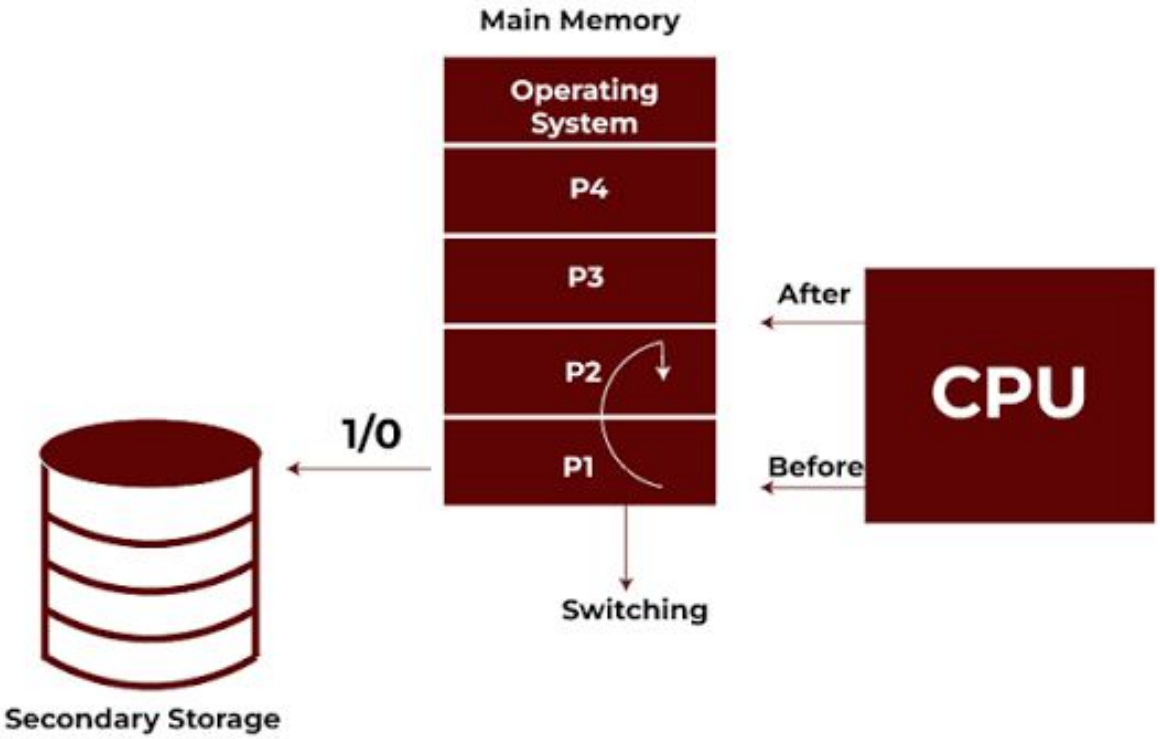
- Better CPU utilization
- Reduced idle time
- Increased efficiency

Example:

Early IBM OS

Memory
Program A
Program B
Program C
↓
CPU executes whichever is ready.

If one program is waiting for data from the hard disk, the CPU immediately starts executing another program. This keeps the processor busy.


Main Memory
Operating System
P4
P3
P2
P1
After
Before
CPU
I/O
Switching
Secondary Storage

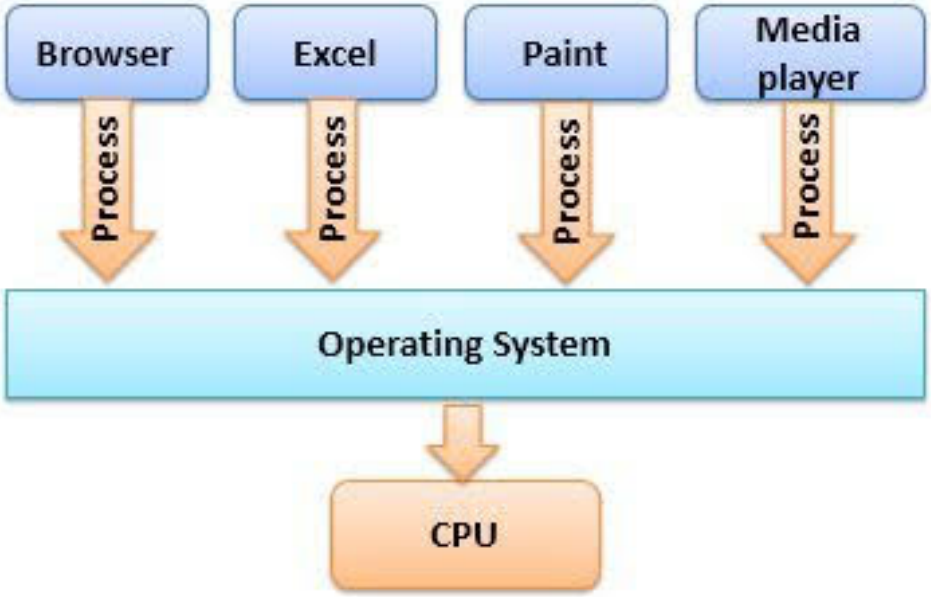
Multitasking Operating System

Definition
Allows a single user to run multiple applications at the same time.

Examples

- Windows
- macOS
- Linux

Real-Life Example
Listening to Spotify while typing in Microsoft Word and downloading a file.



Multitasking

Multiprogramming vs Multitasking

Feature	Multiprogramming	Multitasking
Users	Multiple programs	Single user
Goal	CPU utilization	User convenience
Switching	On I/O wait	Frequent time slices
Example	Early Mainframes	Windows, Linux

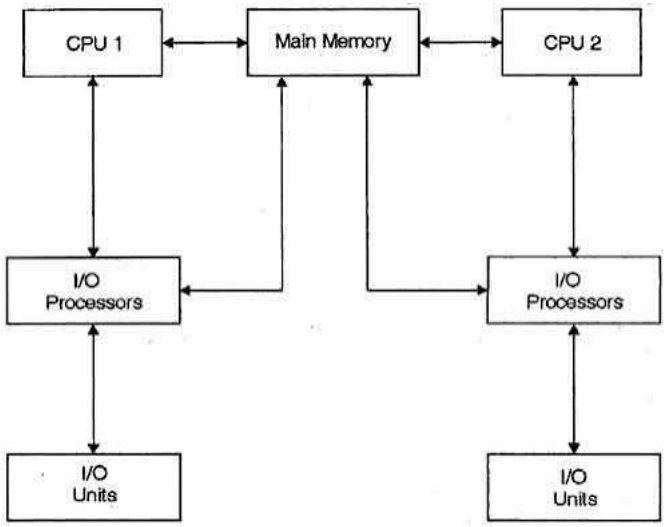
Clarify that both improve CPU efficiency, but multitasking focuses on interactive user experience.

Multiprocessing Operating System

Definition
Uses two or more CPUs or processor cores to execute tasks simultaneously.
Advantages
Faster execution
High reliability
Increased throughput
Examples
Multi-core desktop computers
Servers

Diagram
CPU 1 ← Tasks
CPU 2 ← Tasks
CPU 3 ← Tasks

Today’s laptops often have 8 or more processor cores, enabling true parallel execution.


CPU 1
Main Memory
CPU 2
I/O
Processors
I/O
Processors
I/O
Units
I/O
Units

Time-Sharing Operating System

Definition
Multiple users share the CPU through very short time slices.

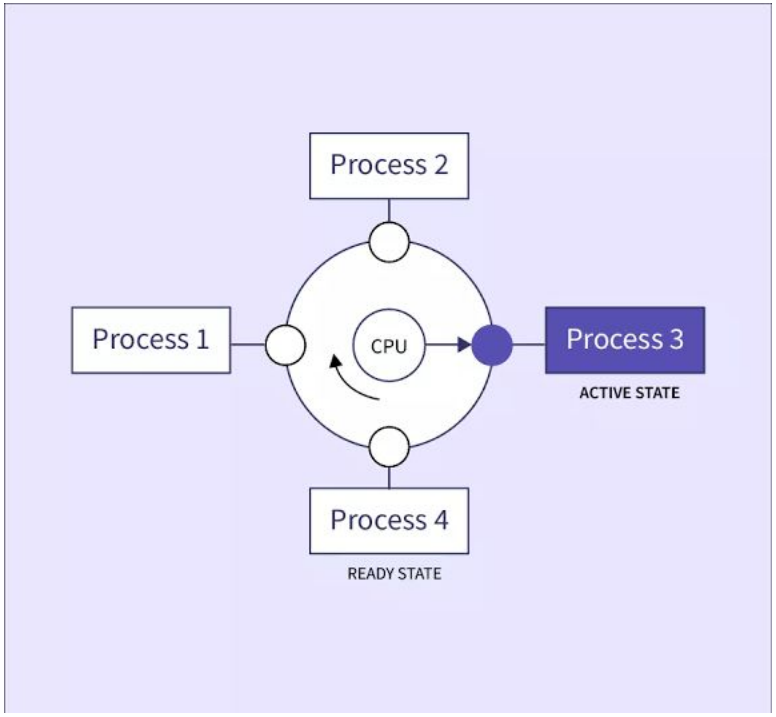
Characteristics

- Interactive
- Fast response
- Fair CPU allocation

Example
UNIX systems

Diagram
User 1 → CPU (10 ms)
User 2 → CPU (10 ms)
User 3 → CPU (10 ms)

Although many users appear to use the system simultaneously, each receives only a tiny slice of CPU time.


Process 2
Process 1
CPU
Process 3
ACTIVE STATE
Process 4
READY STATE

Distributed Operating System

Definition
Several independent computers work together as one integrated system.

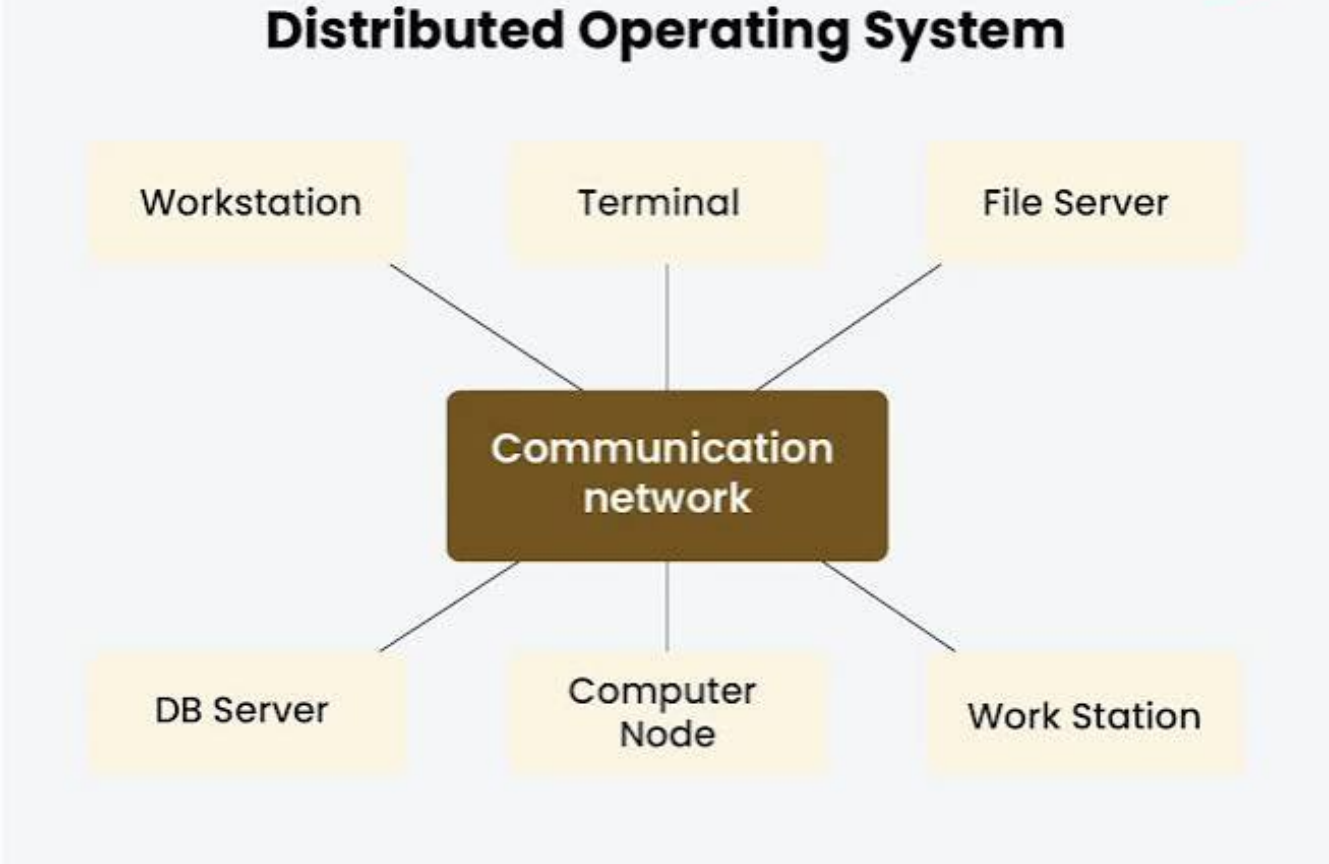
Characteristics

- Resource sharing
- Fault tolerance
- High performance

Examples

- Scientific computing clusters
- Cloud infrastructure

Diagram
Computer 1 ↔ Computer 2 ↔ Computer 3 ↔ Computer 4


Distributed Operating System
Workstation
Terminal
File Server
Communication network
DB Server
Computer Node
Work Station

Network Operating System

Definition

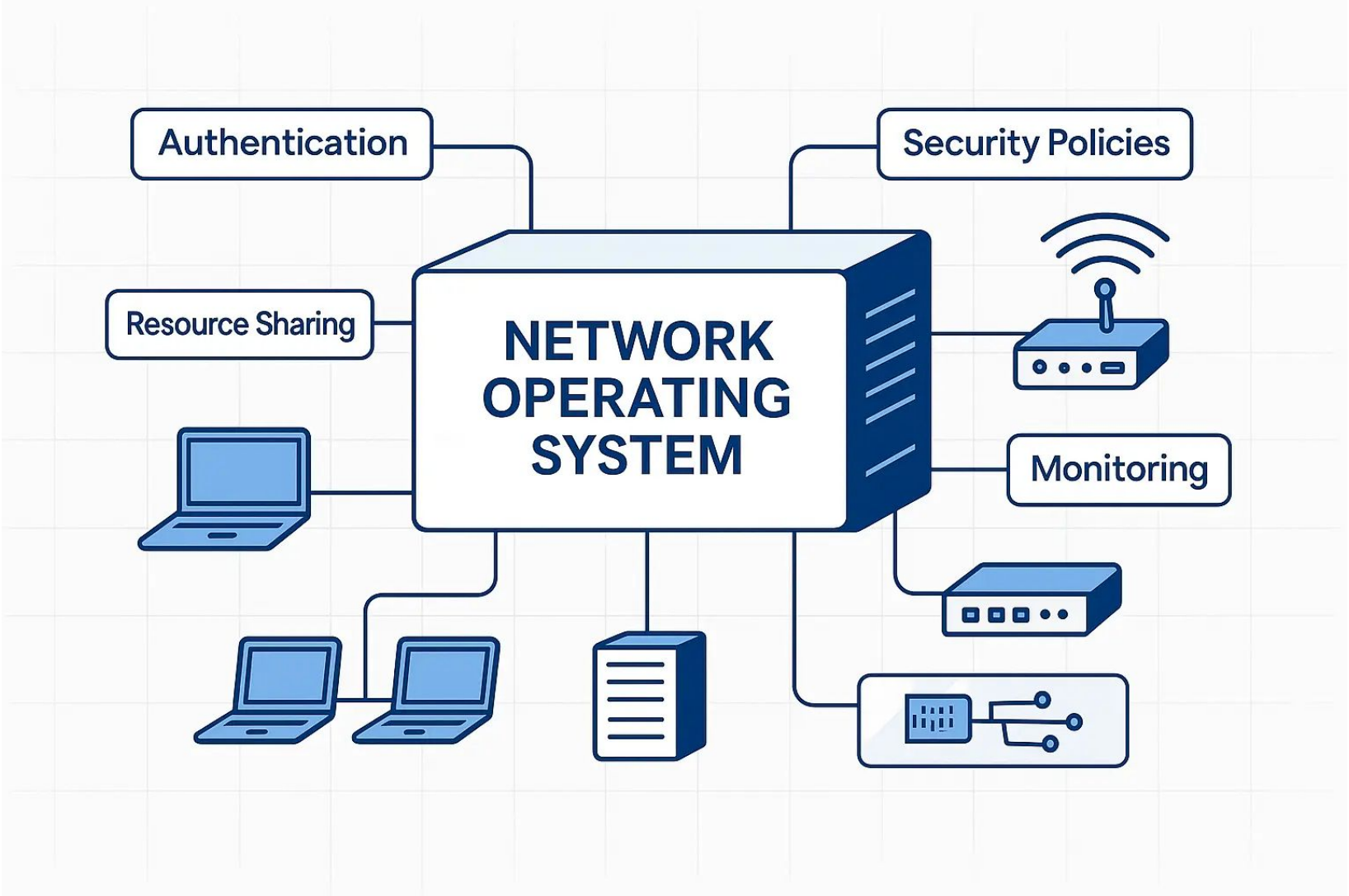
Manages communication and resource sharing among computers connected through a network.

Services

- File sharing
- Printer sharing
- User authentication
- Remote login

Examples

- Windows Server
- Linux Server


Authentication
Security Policies
Resource Sharing
NETWORK OPERATING SYSTEM
Monitoring

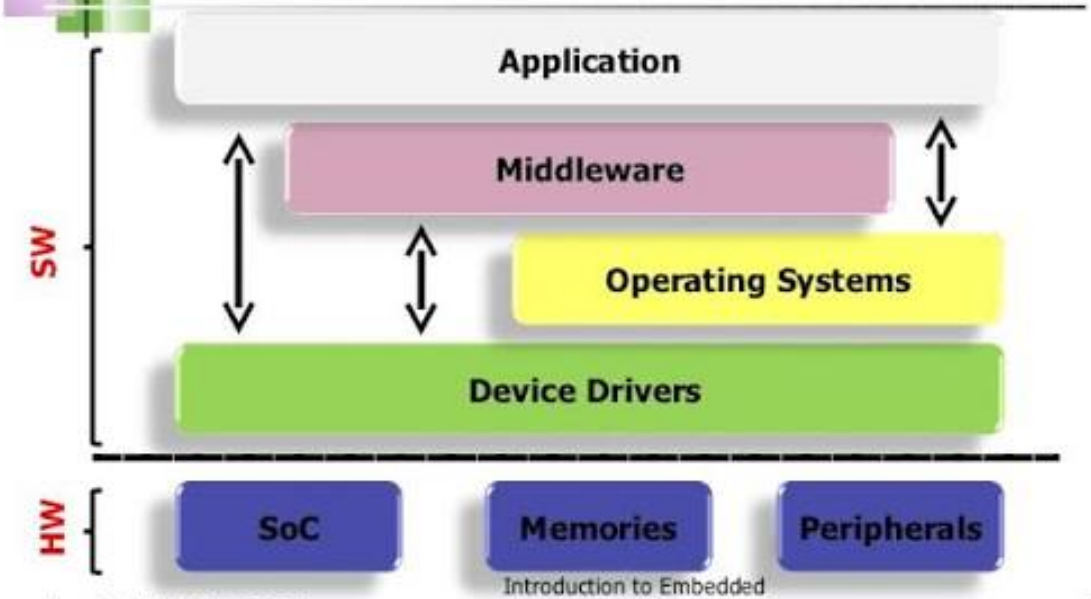
Embedded Operating System

Definition
Designed for dedicated devices with limited memory and processing power.
Examples

- Washing machines
- Microwave ovens
- Smart TVs
- Digital cameras
- Routers

“Unlike your laptop, an embedded operating system performs a single dedicated task efficiently.”

Embedded Systems Architecture



Mobile Operating System

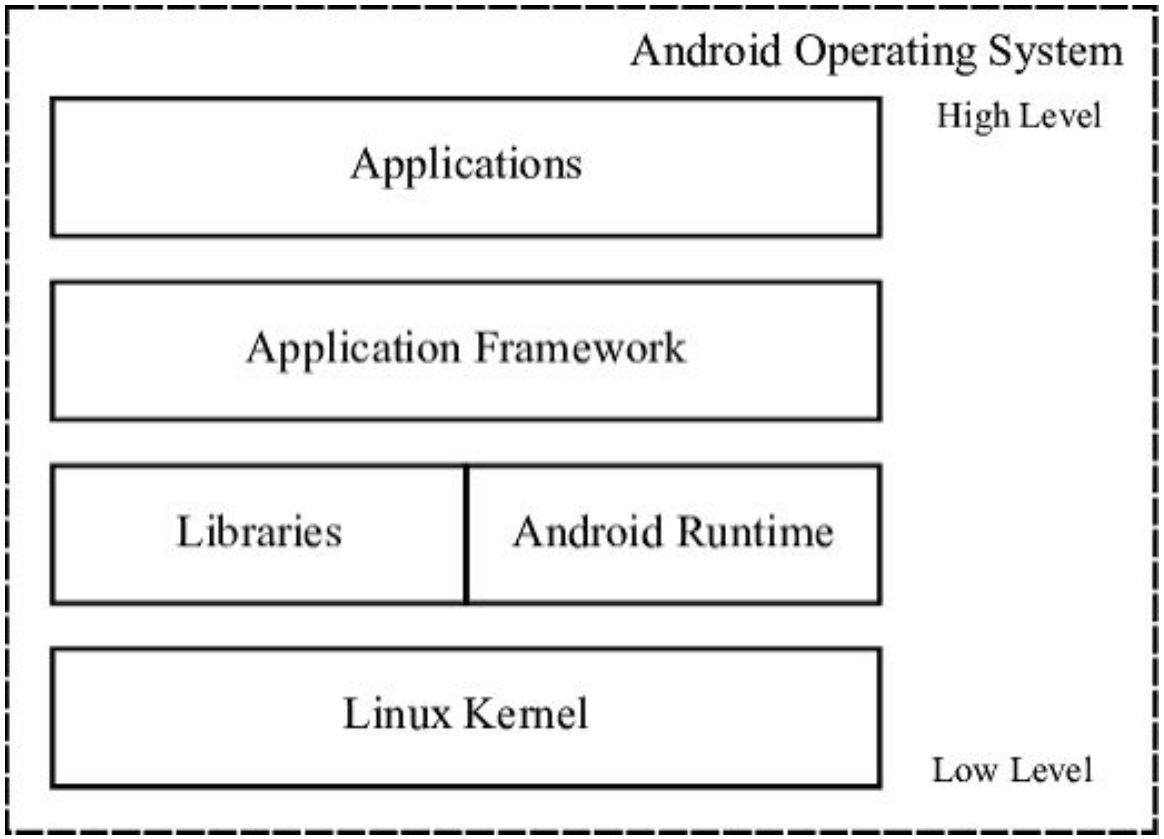
Examples

- Java OS
- Android
- iOS

Features

- Touch interface
- Power management
- App ecosystem
- Wireless communication

Why is Android suitable for smartphones but not for bank servers?


Android Operating System
High Level
Applications
Application Framework
Libraries
Android Runtime
Linux Kernel
Low Level

Comparison of Operating Systems

Type	Users	Example	Application
Batch	Many jobs	IBM	Payroll
Multiprogramming	Multiple programs	Early IBM	Mainframes
Multitasking	Single user	Windows	Personal computers
Multiprocessing	Multiple CPUs	Linux	Servers
Time Sharing	Many users	UNIX	Universities
Distributed	Multiple computers	Linux Cluster	Cloud
Network	Network users	Windows Server	Offices
Embedded	Single purpose	RTOS-based devices	Appliances
Mobile	Mobile users	Android	Smartphones

Which Operating System Is Used Where?

- ATM → Embedded OS
- Laptop → Windows/macOS/Linux
- Mobile → Android/iOS
- Server → Linux/Windows Server
- Smartwatch → Embedded OS
- Data Center → Linux
- University Lab → Linux/Windows

Summary

- Batch OS
- Multiprogramming OS
- Multitasking OS
- Multiprocessing OS
- Time-Sharing OS
- Distributed OS
- Network OS
- Embedded OS
- Mobile OS

Q1. Which type of Operating System processes jobs in groups without direct user interaction?
A) Time-Sharing OS B) Batch OS C) Distributed OS D) Mobile OS

Q2. Which Operating System allows a user to run multiple applications seemingly at the same time?
A) Batch OS B) Embedded OS C) Multitasking OS D) Distributed OS

Q3. Which type of Operating System uses two or more processors/CPU cores to execute tasks?
A) Multiprocessing OS B) Batch OS C) Time-Sharing OS D) Network OS

Q4. Which Operating System is primarily designed for smartphones and tablets?
A)Batch OS B)Mobile OS C)Network OS D)Distributed OS

Q5. Which Operating System enables multiple users to share CPU time through small time slices?
A) Embedded OS B) Batch OS C)Time-Sharing OS D)Multiprocessing OS